

Signal-Aware Framework for Multilingual Language Model Evaluation

1. Research Objective

Training multilingual language models requires billions of tokens and hundreds of thousands of euros in compute. Throughout training (pre-training, mid-training, and post-training), practitioners must repeatedly choose data mixtures and hyperparameters that determine the final model quality. These decisions are typically guided by evaluation on benchmark tasks. However, not all benchmarks provide equally informative signals for training decisions. Benchmarks vary in their diagnostic value; some tasks are sensitive to changes in training data or model configuration, while others exhibit high variance, redundancy, or weak correlation with downstream objectives. Consequently, improvements on certain benchmarks may not reflect meaningful progress for the aspects of model behavior that practitioners seek to optimize.

At the same time, large benchmark suites incur non-trivial evaluation cost, particularly during early and intermediate stages of training when models are evaluated frequently. This creates a practical tradeoff between evaluation coverage: assessing many tasks to capture diverse capabilities, and evaluation efficiency: limiting the computational overhead required to obtain feedback during training. Despite the central role of benchmarks in guiding model development, it remains unclear which benchmarks provide reliable and informative signals for training decisions across different stages of multilingual model training. As a result, practitioners often evaluate on extensive benchmark suites with diminishing diagnostic returns, or rely on small subsets of tasks whose signals may be noisy or misaligned with the decisions being made. Systematically identifying benchmarks that provide meaningful, stage-specific diagnostic signal could substantially improve the efficiency and reliability of multilingual model development.

Recent work has begun to examine the diagnostic value and efficiency of language model benchmarks, recognizing that not all benchmarks provide equally informative signals for guiding model development. Heineman et al. (2025) introduce a signal-to-noise ratio (SNR) framework that quantifies benchmark reliability for English models. In this framework, signal measures how well a benchmark separates models of similar scale, while noise captures variability in scores across training checkpoints or stochastic runs. Their empirical analysis shows that benchmarks with higher SNR more consistently preserve model rankings across scales, improving the accuracy of training decisions and reducing error when extrapolating performance using scaling laws. This work highlights that benchmark choice directly affects the reliability of decisions made during model development.

Other work focuses on improving evaluation efficiency without sacrificing diagnostic signal. Chen et al. (2024) study the scaling behavior of downstream tasks and find that some benchmarks follow predictable scaling trends while others do not, suggesting that only a

subset of tasks may be necessary to forecast final model performance [2]. Gupta et al. (2024) propose SMART filtering, which removes contaminated, redundant, or overly easy examples from benchmark datasets [3]. Their results show that filtering such examples can substantially reduce evaluation cost while preserving agreement with independent measures of model quality, indicating that many benchmark items contribute little useful signal. Similarly, Zhou et al. (2025) analyze benchmark reliability through item response theory, demonstrating that many widely used benchmarks contain items with weak discriminative power and that smaller, carefully selected subsets can produce more stable and interpretable model comparisons [4].

Together, these works suggest that benchmark effectiveness depends not only on coverage but also on how much reliable signal a benchmark provides relative to its evaluation cost. However, existing studies primarily focus on English benchmarks and general model ranking. ***Our goal is to extend the aforementioned frameworks to multilingual settings*** and identify the most reliable multilingual benchmarks for each training stage that correlate with final model performance. This will enable practitioners to make better, cheaper decisions during multilingual model development.

2. Research Plan

Timeline	Task
April	Model training
April	Signal-aware framework implementation
April	Benchmark and model selection
May	Model evaluation and SNR analysis
May	Multicultural regional knowledge benchmark analysis
June - July	Framework extension across design decisions and benchmarks
August	Validation on Apertus
September	Dissemination

Phase 1: Model Training

- Pretrain 36 small models (4 model sizes ranging from 175M to 1B params) on 3 multilingual data mixtures from FineWeb (English data) and FineWeb2 (multilingual data covering more than 200 languages), with 3 seeds.

Phase 2: Signal-Aware Framework Implementation

- Implement the signal-aware framework, including an optimised evaluation strategy, and calculations of the signal, noise, SNR, decision accuracy, and scaling-law error.

Phase 3: Benchmark and Model Selection

- Model Suite: For each size-mixture pair of the small custom models, we will select the best-performing model among the three seeds based on the loss. Besides these 12 models, we will evaluate three families of open-source multilingual models (Apertus, OLMo and SmoLLM) with available intermediate checkpoints and sizes ranging from 1B to 70B.
- Benchmark Suite: We will evaluate 40 multilingual benchmarks, including cross-lingual understanding (XNLI, XCOPA, XStoryCloze, Belebele), QA (XQuAD), reasoning (MGSM, XLSum), regional knowledge (INCLUDE, Global MMLU), and instruction-following (IFEval). We will use logprob evaluation and consider different evaluation setups (0-shot vs. 5-shot).

Phase 4: Model Evaluation and SNR Analysis

For each benchmark, we will compute the signal, noise, SNR, decision accuracy, and scaling-law error. We will analyze results to identify which benchmarks and subtasks have high SNR at each stage, enabling stage-specific evaluation strategies.

Phase 5: Multicultural Regional Knowledge Benchmark Analysis

We will conduct a thorough analysis of INCLUDE [5], our internally developed benchmark designed to evaluate regional knowledge across more than 100 countries. Using this analysis, we will identify optimal benchmark subsets that provide reliable evaluation signals for multilingual models at different stages of training.

Phase 6: Framework Extension across Design Decisions and Benchmarks

Extend the framework along three directions:

- Design decisions: Study model design decisions beyond data mixtures, such as tokenizer choice, by training and evaluating additional models under controlled conditions.
- Benchmark taxonomy: Expand the analysis of high-signal multilingual benchmarks by grouping them not only by training stage, but also by capabilities evaluated, language family, and script.
- Benchmark design recommendations: Explore how the framework can inform benchmark design itself, generating guidelines for constructing high-quality multilingual benchmarks.

Phase 7: Validation on Apertus

We will validate our findings during the training of the next version of Apertus, our multilingual model, where we will have access to all training checkpoints and can measure the real-world impact of benchmark-driven decisions.

Phase 8: Dissemination

Deliverables:

- Paper: Submit a paper with detailed findings, methodology, SNR scores and stage-specific recommendations for all evaluated multilingual benchmarks.
- Open data: Share the optimal benchmark subsets per analysis axis.
- Evaluation toolkit: Open-source Python code for computing SNR on custom benchmarks and models.

3. Computational Budget

We estimate the total computational budget at 100,000 GPU hours, accounting for the evaluation of 40 benchmarks in 0- and 5-shot settings across 25 checkpoints for 50 models ranging from 175M to 70B parameters.

References

- [1] Heineman, David, Valentin Hofmann, Ian Magnusson, Yuling Gu, Noah A. Smith, Hannaneh Hajishirzi, Kyle Lo, and Jesse Dodge. "Signal and noise: A framework for reducing uncertainty in language model evaluation." <https://arxiv.org/pdf/2508.13144> (2025).
- [2] Chen, Yangyi, Binxuan Huang, Yifan Gao, Zhengyang Wang, Jingfeng Yang, and Heng Ji. "Scaling laws for predicting downstream performance in llms." arXiv preprint <https://arxiv.org/abs/2410.08527> (2024).
- [3] Gupta, Vipul, Candace Ross, David Pantoja, Rebecca J. Passonneau, Megan Ung, and Adina Williams. "Improving model evaluation using smart filtering of benchmark datasets." In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies <https://aclanthology.org/2025.naacl-long.235/> (2025).
- [4] Zhou, Hongli, Hui Huang, Ziqing Zhao, Lvyuan Han, Huicheng Wang, Kehai Chen, Muyun Yang et al. "Lost in benchmarks? rethinking large language model benchmarking with item response theory." <https://arxiv.org/abs/2505.15055> (2025).
- [5] Romanou, Angelika, Negar Foroutan, Anna Sotnikova, Zeming Chen, Sree Harsha Nelaturu, Shivalika Singh, Rishabh Maheshwary et al. "Include: Evaluating multilingual language understanding with regional knowledge." arXiv preprint <https://arxiv.org/abs/2411.19799> (2024).
- [6] Hernández-Cano, Alejandro, Alexander Hägele, Allen Hao Huang, Angelika Romanou, Antoni-Joan Solergibert, Barna Pasztor, Bettina Messmer et al. "Apertus: Democratizing open and compliant llms for global language environments." arXiv e-prints <https://arxiv.org/abs/2509.14233> (2025).